

Overview

The World Tennis Number (WTN) is a rating system underpinned by the Glicko-2 algorithm [1]. The intention is to have every tennis player in the world on a common scale, ranging from 1 (best) to 40 (worst).

The aim is to accurately rate players, regardless of their background, e.g. their sex, age, nationality, etc. This presents a challenge since players play most of their matches in homogeneous groups, often determined by their background, with only a few matches being played between different groups.

A hierarchical approach has been developed at Imperial, which assigns ratings to groups, as well as players.

Glicko

The Glicko algorithm [2] is an approximation to a Bayesian model that updates player skill. The model is closely related to the Bradley-Terry model for paired comparisons [3], since the probability for game outcomes is

$$Pr(s_{ijk} = s) = \frac{(10^{(\theta_i^{(t)} - \theta_j^{(t)})/400})^s}{1 + 10^{(\theta_i^{(t)} - \theta_j^{(t)})/400}}$$

where $s_{ijk} = 1$ when player i beats player j and $s_{ijk} = 0$ when player j beats player i in the k^{th} game, and each player's skill is,

$$\theta^{(t+1)} \sim \mathcal{N}(\theta^{(t)}, \sigma^2)$$

To update a player's skill, we take their prior skill and their game outcomes and approximate the marginal posterior distribution of their skill. This is determined by integrating out the opponents' strength parameters over their prior distribution rather than their posterior distribution. So if a player plays n_j games against m opponents, then their marginal posterior distribution can be written as

$$f(\theta|s) \approx \phi(\theta) \prod_{j=1}^m \int \prod_{k=1}^{n_j} Pr(s_{ijk}) \phi(\theta_j) d\theta_j$$

Glicko-2

Glicko-2 is a variant of the original Glicko algorithm. It introduced a volatility parameter that allows for stochastic variance. The model assumes that each player's skill evolves as follows,

$$\log \sigma^{2(t+1)} \sim \mathcal{N}(\log \sigma^{2(t)}, \tau^2)$$

$$\theta^{(t+1)} \sim \mathcal{N}(\theta^{(t)}, \sigma^{2(t+1)})$$

Doubles Tennis

The original Glicko-2 was intended to rate teams. However, in doubles tennis players often switch partners, so we would prefer to rate the individuals rather than the pairs.

The idea of a 'notional opponent' was introduced to distill the individuals' ratings from the pairs ratings. Instead of 1 game between 2 pairs, we create 4 games for each individual involved in a match against their respective 'notional opponent'.

Notional Opponent

Consider a player with skill θ , their notional opponent's skill is a function of the skill of all the players involved:

$$\theta_{\text{Notional}} = f(\theta_{\text{Opp1}}, \theta_{\text{Opp2}}) + \theta - f(\theta, \theta_{\text{Partner}})$$

where $f(\cdot, \cdot)$ is a skill aggregation function.

Some skill aggregation functions proposed include:

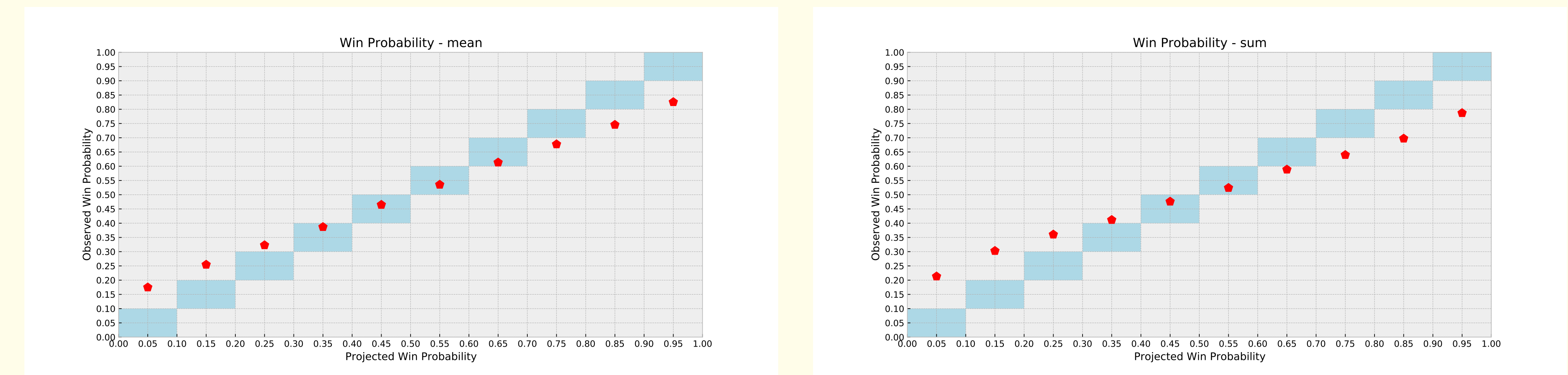
- Sum - $f(\theta_i, \theta_j) = \theta_i + \theta_j$
- Mean - $f(\theta_i, \theta_j) = \frac{1}{2}(\theta_i + \theta_j)$
- Max - $f(\theta_i, \theta_j) = \max(\theta_i, \theta_j)$
- Min - $f(\theta_i, \theta_j) = \min(\theta_i, \theta_j)$
- Geometric - $f(\theta_i, \theta_j) = \sqrt{|\theta_i| |\theta_j|}$

Skill Aggregation Functions

The skill aggregation functions were compared using tennis match data collected by the ITF from 2016 to 2020.

ITF Data	Sum	Mean	Max	Min	Geometric
Upset Ratio	34.68%	34.16%	35.11%	34.28%	34.16%
ATP Top 500	$\frac{179}{500}$	$\frac{167}{500}$	$\frac{161}{500}$	$\frac{160}{500}$	$\frac{167}{500}$

Upset Ratio = $\frac{\text{Unexpected Wins}}{\text{Total Games}}$, ATP Top 500 = No. players ranked in the top 500 by the ATP and WTN. Below is a breakdown of the projected probabilities for the mean (left) and sum (right) aggregation functions:



The top 5 players, using the mean aggregation function, are:

Player	Rank	Rating	Deviation	Volatility	ATP Rank	W/L
Rafael Nadal	1	4068.72	130.94	0.06019	424	13/5
Daniel Lustig	2	3765.41	183.30	0.06007	1672	5/3
Jack Sock	3	3740.69	48.06	0.06148	118	110/49
Mikael Ymer	4	3634.72	114.21	0.06004	1055	8/9
Alexander Zverev	5	3628.11	64.22	0.06023	97	36/35

Future Work

Comparing our rankings with the ATP rankings highlights the need for a penalty on inactivity. Many highly ranked players have large rating deviations due to few matches played. The next step in improving the rating system is to implement a shrinkage term that will pull players downward to the mean rating of their group if they do not play during a rating period.

Currently the grouping structure for the doubles tennis is based on the grade and circuit of the match. This could be improved upon by using a community detection algorithm, e.g. Leiden algorithm.

References

- [1] Mark E Glickman. "Dynamic paired comparison models with stochastic variances". In: *Journal of Applied Statistics* 28.6 (2001), pp. 673–689.
- [2] Mark E Glickman. "Parameter estimation in large dynamic paired comparison experiments". In: *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 48.3 (1999), pp. 377–394.
- [3] Ralph Allan Bradley and Milton E Terry. "Rank analysis of incomplete block designs: I. The method of paired comparisons". In: *Biometrika* 39.3/4 (1952), pp. 324–345.